

What a Motor Prior Is Worth: A Body Model and a Teacher Compared to Convergence and Through Withdrawal

Anonymous Author(s)

Abstract—The guidance hypothesis holds that guidance breeds dependence: the longer a learner is guided, the larger its fall when guidance is withdrawn. We test it, and the same question for a model of the body, in one soft actor-critic learner on six muscle-driven bodies, two from MyoSuite and four planar arms of one to four joints, given a frozen forward model learned from reward-free babbling or a frozen teacher acting through a distillation term, with matched seeds, an adversarial control for each prior and training to a checked plateau. Replacing a withdrawn teacher by a frozen snapshot of the student, which keeps the loss term but adds no information, separates what withdrawal removes: the teacher’s share of the immediate drop falls with how long it was attached, the deleted term’s does not. Once training has converged, a learner that lost its teacher is never detectably worse than one never guided, at any attachment duration, on either body, whether guidance supplemented or replaced its own gradient; with a competent teacher it is better, by more the longer it was guided. A withdrawn body model can leave the learner short of one that kept it, never behind one that never had it. What remains after withdrawal is a shortfall against keeping the prior, not dependence. What the priors are worth is bound to their content: the babble model beats model-free learning on four of six bodies and a random model on all six, and a teacher is worth its competence.

Index Terms—Developmental robotics, guidance hypothesis, internal models, musculoskeletal control, reinforcement learning.

I. INTRODUCTION

HUMANS and animals acquire motor skills in a handful of trials; deep reinforcement learning on muscle-actuated bodies can need hundreds of millions of environment steps. Computational sensorimotor theory offers two accounts of what a biological learner brings to a task that an agent trained from scratch does not. One is an internal forward model of the body, predicting the sensory consequences of motor commands [1], [2] and adapted in the intrinsic coordinates of joints and muscles [3]. The other is guidance: the guidance hypothesis [4], [5], [6] holds that augmented feedback steers the learner toward the correct response while breeding a dependence on it, so that the longer it is given the larger the fall when it is withdrawn. Both are claims about a *prior* in the developmental sense, something present before task learning begins, and both can be instantiated for an artificial learner, compared on equal terms, and, what human studies rarely do, taken away in pieces.

We do that here. Under one soft actor-critic backbone on six muscle-driven bodies, we add either a frozen forward model of the body, learned once from reward-free motor babbling and

used for short-horizon imagined rollouts, or a frozen teacher policy acting through a distillation term. Each route is given the adversarial control the other has: a randomly initialised model in the same imagination machinery, a randomly initialised teacher in the same distillation loss. Every condition has at least twelve matched seeds, every comparison is a paired contrast on the seed, and training runs until the learning curves have plateaued. We propose no algorithm; like other simulation studies of what an endowment is worth to a learner [7] or of what removing a contingency does [8], the claim is about what withdrawing a motor prior costs, and secondarily about what the prior was worth.

The withdrawal cost decomposes, and what is left at convergence is not dependence. Zeroing a teacher’s distillation weight deletes a loss term as well as the teacher’s information. Replacing the teacher by a frozen snapshot of the student keeps the term but carries nothing further, and shows on both MyoSuite bodies that the teacher’s share of the immediate drop falls with attachment while the deleted term’s does not; where the drop grows with attachment, the term is what grows. Measured as the literature measures dependence, the withdrawn learner against one never guided, no withdrawal leaves a deficit once training has converged, and with a competent teacher the withdrawn learner is ahead. The body model behaves alike once its withdrawal is made single-factor: purging the imagined buffer also doubles the real data per update, and with that held fixed the loss is detected on the elbow at 12k (+22 mrad), where the standard protocol detects nothing, and is not detected at 100k, while on the finger it grows to +15 mm, the imagination forgone after withdrawal. On neither body is the withdrawn learner worse than one that never had a model.

What the priors are worth is smaller and bound to their content. At convergence the babble-trained body model beats model-free learning on four of the six bodies and is never detectably worse; it beats a random model on all six, and the random model’s harm grows with joint count. A teacher is worth its competence: on the finger a teacher within 3.2 mm of doing nothing caps the student 28 mm below model-free learning, and a competent teacher on the same body makes the student better than every condition without it.

Every run of an earlier, unpublished 12k version of this study that has a counterpart here is reproduced seed for seed; scoring at that fixed endpoint would have changed what the body-model results detect on both MyoSuite bodies. Code and data will be released on acceptance.

II. BACKGROUND AND RELATED WORK

The guidance hypothesis. Salmoni, Schmidt, and Walter [4] argued that frequent augmented feedback guides performance during acquisition but degrades learning; the account was named the guidance hypothesis [9] and generalised to practice conditions [5]. High-frequency physical guidance gave the poorest retention in one test [6] and reduced knowledge of results improved delayed retention in another [10], but a preregistered meta-analysis finds no reliable benefit of reduced feedback frequency [11], [12]. The hypothesis is a claim about *dependence*, the learner being worse after withdrawal than one never guided or guided less, and its mechanism is that guidance *substitutes* for the learner’s own error correction. Human studies have kept a feedback channel while corrupting or removing its information [13], [14], [15], but none separates the information from the channel as guidance is withdrawn. Computational models of dependence on robotic assistance place it in an effort or slacking term of the learner’s own cost [16], [17].

Teacher–student learning. Privileged learning [18], [19] distils a teacher that sees state the deployed policy will not into a student on the student’s own rollouts [19], [20]; RMA [21] regresses the privileged latent instead. Distillation may be the student’s entire training, or an auxiliary term annealed or cut to make the student independent [22], [23]; weaning can collapse the student, which Agarwal et al. [23] attribute, without a test, to the change of objective. A term toward the learner’s own earlier policy stabilises fine-tuning [24], [25]; supervised distillation uses the student itself or a random network as the teacher to separate what the term does from what the teacher knows [26], [27], and compares stopping distillation mid-way only with continuing it [28]; in language models a teacher-free signal can match a noisy teacher [29]. To our knowledge no study splits the cost of withdrawing a teacher into lost information and a deleted loss term by retargeting the term to the learner’s own frozen snapshot, which is formally a trust region to an old policy, used here as a counterfactual control. A cloned policy fine-tuned by off-policy reinforcement often degrades at the transition, because a fresh or miscalibrated critic gives unreliable values outside the data [30], [31], [32]; a retained cloning term is reported to prevent this [33].

Developmental robotics. A body model built from self-generated movement and a caregiver’s guidance are the two routes this study instantiates. Developmental robotics [34], [35], [36] treats motor babbling as a way to learn forward models of the self [37], intrinsic motivation as what drives it [38], and infant reaching as the paradigm case [39]. Scaffolding theory counts a tutor as successful when the learner does the task alone after the tutor fades [40]; the withdrawn learner against one never guided, at convergence, is that criterion measured.

Model-based reinforcement learning (RL) and body models. World Models [41] trains a model once from random rollouts; the Dreamer series [42], [43], [44] and TD-MPC2 [45] keep the model learning online; MBPO [46] motivates the short branched rollouts of our Dyna step. Self-modelling [47] and goal babbling [48] learn body models by directed

exploration; Plan2Explore [49] and APV [50] pretrain a model without task reward and reuse or fine-tune it; offline methods [51], [52], [53] fit a model to logged, rewarded data and optimise conservatively under it. Most muscle RL is model-free [54], [55], [56], [57]; muscle world models are co-learned with the skill [58], [59], and frozen models learned from reward-free actuation already drive soft and muscle-driven bodies by planning or model-only RL [60], [61], [62]. We use such a model only for short Dyna rollouts mixed with real data, price it at convergence, and control it with a random model; that an inaccurate model can make model-based RL worse than model-free, more so in higher dimension, is known [46], [63]. MyoSuite [64] provides two of the bodies.

III. METHOD

A. Bodies, task and backbone

`myoElbowPose1D6M` is a one-degree-of-freedom elbow driven by six muscles; error is a joint-angle error in milliradians (mrad). `myoFingerReachRandom-v0` is a four-joint index finger driven by five muscles reaching a three-dimensional fingertip target; error is a Euclidean fingertip error in millimetres (mm). Four planar arms of one, two, three and four revolute joints, driven by two, four, eight and ten MuJoCo muscles (antagonist pairs, biarticular from three joints on), reach a planar fingertip target; error is in mm. Errors are never compared across bodies. The task is reach; episodes are 100 steps from a fixed start pose; training draws from eight fixed targets; evaluation is greedy on twenty disjoint held-out targets, so every number here is a generalisation measurement (on the one-joint bodies an interpolation test; Sec. VI). One soft actor-critic (SAC) [65], [66] backbone is used in every condition: two hidden layers of 128 units, update-to-data ratio 2, batch 128, 1,000 warm-up steps, learning rate 3×10^{-4} , automatic entropy temperature. Conditions differ only in what is added.

B. The two priors

The *forward-model* route adds a body model $f_\theta(s, a) \mapsto s'$ trained once on task-agnostic, reward-free Ornstein–Uhlenbeck muscle activations (10^5 transitions on the elbow, 2×10^5 on the finger and on each planar arm), then frozen, and used for short-horizon Dyna [67] rollouts branched from sampled real states; the environment’s analytic reward is evaluated at the imagined states, so every reading of this route is scoped to a forward model *plus* a known reward function. Each update draws 64 real and 64 imagined transitions where model-free conditions draw 128 real.

The *guidance* route adds a frozen teacher actor. The student’s actor loss is

$$\mathcal{L}_\pi = q \mathcal{L}_{\text{SAC}} + c \left\| \tanh \mu_{\text{student}}(o) - \tanh \mu_{\text{teacher}}(o) \right\|^2, \quad (1)$$

on replayed observations o , with $c = 1$ while the teacher is attached; the critic trains on reward in every condition. In the *additive* regime ($q = 1$ always) the term supplements the student’s own policy gradient, as concurrent augmented feedback does; this is the coach used everywhere unless stated. In the *replacing* regime ($q = 0$ while attached, $q = 1$

from withdrawal) the actor is moved only by the teacher until withdrawal, as physical guidance moves the limb and as reward-free distillation [19], [20] does.

Three teachers are used. T1 on each MyoSuite body is the terminal actor of a model-free run of 30k rewarded steps on a seed outside the study's twelve; it scores 27.2 mrad on the elbow's held-out targets, and 169.1 mm on the finger's, where a zero-activation policy scores 172.3, so the finger T1 is within 3.2 mm of doing nothing. T2 on the finger is the terminal actor of a body-model run of 100k steps on seed 1 of the study (35.6 mm held-out); it is used for every finger withdrawal experiment and reported beside T1 in the core grid. T2 varies teacher quality and is not budget-matched to anything. One of the twelve students shares its initialisation seed with the run that produced T2; excluding that student changes no finger-T2 conclusion (every contrast keeps its sign and detection).

C. Conditions

Names below are used in every table. **Core conditions.** `blank` is the model-free backbone alone; `blank64` is the backbone with every update built from 64 distinct real transitions, each repeated once to fill the batch, the real-data content of a Dyna update and no model; with a mean-reduced loss this is SAC at batch 64; `prior` adds the frozen body model; `coach` adds the frozen teacher in the additive regime. **Adversarial controls.** `randprior` replaces the babble-trained model with a randomly initialised frozen model in identical Dyna machinery; `randcoach` replaces the trained teacher with a randomly initialised one distilled through the same loss at the same weight. `priorcoach` carries both aids.

Withdrawing the teacher at step t_w : `none` (no teacher, identical to `blank`); `constant` ($c = 1$ throughout, identical to `coach`); `abrupt` ($c \rightarrow 0$ at t_w , the literature's withdrawal); `selfanchor` (at t_w the teacher is replaced by a frozen copy of the student's own actor, c unchanged: the term survives with the same weight and form but carries no external information); `randanchor` (replaced by a freshly initialised actor); and, in the replacing regime only, `gateon` ($q \rightarrow 1$ at t_w with the real teacher kept). The literature's measure of dependence is `abrupt` – `none`, the withdrawn learner against one never guided. Three paired contrasts attribute the cost: `abrupt` – `constant` is losing the teacher *and* the loss term (the total), `selfanchor` – `constant` is losing the teacher's information only (the teacher component), and `abrupt` – `selfanchor` is losing the loss term only (the loss-term component). The snapshot holds whatever of the teacher the student had absorbed by t_w , so the teacher component prices the teacher's *further* information. Since `none` and `constant` are shared across t_w , `abrupt` – `none` and the total have the same trend with attachment. Every withdrawal is a single step.

Withdrawing the body model at step 8000: `keep` never withdraws (identical to `prior`); `soft` stops generating imagined transitions but leaves the imagined buffer in place, so updates `keep` drawing 64 real and 64 stale imagined; `purge` also empties the buffer, after which every batch is 128 real, the strict withdrawal, which removes imagination and doubles distinct real transitions per update at once; `matched` withdraws as `purge` does but draws 64 real transitions and repeats

them, holding distinct real transitions per update at 64. Only contrasts among `keep`, `soft` and `matched` are single-factor.

D. Metrics and statistics

Endpoint: held-out error at 100k. **Dip:** error at $t_w + 1000$ minus error at t_w . **Attachment trend:** per-seed ordinary-least-squares slope of a component against t_w , then a paired- t interval across seeds, per 1,000 attached steps. The seed is the unit of analysis; every condition has $n = 12$ matched seeds except the three- and four-joint arms, which have $n = 24$; every comparison is a direct paired contrast on the seed, never an inference from two separate verdicts; intervals are paired- t 95% intervals with $df = n - 1$; an interval excluding zero is called significant and marked * or set in bold; a non-significant cell is a failure to detect at this power, never evidence of equivalence. Three endpoints were fixed in advance for the converged grid, 12k, 50k and 100k; counts over the fifty 2k-checkpoints are reported both uncorrected and after a Holm correction over the fifty tests, and are descriptive. Every significant contrast in Tables I and III was also checked with an exact Wilcoxon signed-rank test; the two it does not confirm at $p < 0.05$ are flagged where they appear. Tables II, IV, V and VI rest on the t interval. Convergence is checked per condition as the paired change over the last 20,000 steps; of the 97 conditions in the tables, five have an interval excluding zero and are named where they occur.

E. Implementation details

Actor and critics are two-layer 128-unit SiLU networks, the twin critics with layer normalisation; $\gamma = 0.99$, Polyak $\tau = 0.005$, Adam at 3×10^{-4} , target entropy $-0.5|A|$, gradient norm clipped at 10, real replay 200,000 transitions, imagined replay 400,000, 1,000 warm-up steps, update-to-data ratio 2, batch 128. Dyna: every 100 environment steps, 200 start states are drawn from real replay and rolled $k = 3$ steps under the current actor through the frozen model, with the environment's analytic reward evaluated at the imagined states. The forward model is a two-layer 256-unit SiLU network predicting the next state, trained with Adam at 10^{-3} for 40 epochs on babble data collected with Ornstein-Uhlenbeck activations ($\theta = 0.15$, $\sigma = 0.4$, clipped to $[-1, 1]$, pose re-randomised every 100 steps, no reward). Teachers have the actor's architecture. Evaluation is one greedy 100-step episode on each of the twenty held-out targets, every 2,000 steps, and every 1,000 in the withdrawal conditions. A matched seed sets the network initialisation, replay sampling, the training environment and the order of target draws; the targets are fixed across seeds, and runs are deterministic on CPU. The 1,500 runs behind the tables took about 1,500 single-thread CPU-hours.

F. Reproduction

An earlier, unpublished version of this study scored every condition at 12k with the same code. Every one of its runs with a counterpart here is reproduced to the digit at the shared checkpoints (core grid, both body-model withdrawals, elbow

attachment sweep, replacing regime, anchor conditions); dips that need a 1k evaluation grid are read from these reproduced runs, and the budget check uses the 12k runs as they are.

IV. RESULTS

A. What the priors are worth at convergence

Table I and Fig. 1 score the core conditions on the two MyoSuite bodies at 100,000 steps; every condition in the table has plateaued by the paired-change criterion except the elbow's random-model control, still falling at $-24.99 [-43.78, -6.19]$.

The body model. On the elbow its advantage over model-free learning, $-20.72 [-35.85, -5.59]$ mrad at 12k, is $-4.09 [-11.11, +2.94]$ at 100k, not detected; it is significant at seven of fifty checkpoints uncorrected and one after the Holm correction. On the finger the advantage, not detected at 12k ($-27.10 [-66.32, +12.12]$ mm), is $-40.29 [-57.95, -22.64]$ at 100k and is detected at every checkpoint from 14k on (45 of 50 uncorrected, 42 corrected). The `blank64` condition prices the real-data half of the Dyna update: halving the real transitions per update costs the elbow $+9.68 [-3.69, +23.06]$, not detected, and *helps* the finger by $-14.59 [-26.00, -3.18]$, as smaller batches are known to help off-policy learners by reducing critic overfitting [68], [69], [70]. Against this matched control, and against a random model, the babble model keeps its advantage (Table I).

Table II and Fig. 3 extend the body-model contrasts to four planar arms of one to four joints, with twenty-four seeds on the two hardest. The model beats model-free learning on the one-, two- and four-joint arms and is not detected on the three-joint arm; it beats the random model on every arm, and the random model's harm relative to no model grows monotonically with joint count, from not detected on one joint to $+257.72 [+177.95, +337.48]$ mm on four (per-seed slope of the relative harm, $+0.41 [+0.18, +0.63]$ of the model-free error per joint, exact Wilcoxon $p = 0.003$). The model's own relative advantage does not grow with joint count detectably (slope $-0.10 [-0.25, +0.05]$ per joint, $p = 0.18$). Against `blank64` it is detected on the one-, two- and four-joint arms, though the four-joint `blank64` condition again sits below `blank` ($-62.42 [-129.04, +4.20]$). One condition in the table is still moving at 100k, the one-joint `randprior`, rising ($+13.80 [+3.54, +24.06]$).

The teacher. On the elbow the coached student keeps a small advantage over model-free learning at 100k, $-8.06 [-15.83, -0.28]$, detected by the t interval at 43 of 50 checkpoints uncorrected and 13 after correction, but with an exact Wilcoxon $p = 0.052$ at the endpoint; its lead over the body model, $-10.89 [-19.26, -2.52]$ at 12k, is $-3.97 [-8.47, +0.53]$ at 100k. On the finger the teacher decides everything. With T1, a student held near a policy that barely moves does not improve past 12k (122.77 mm at 12k, 123.00 at 100k), ends $+28.33 [+16.11, +40.56]$ worse than model-free learning, $+68.63 [+59.19, +78.07]$ worse than the body model, and $+13.86 [+7.62, +20.09]$ worse than a student held near a random teacher. With T2 on the same body and seeds the coached student ends at 34.50 mm, level with T2 plus the body model and better than every condition without

T2 (Table I), within 1.1 mm of its teacher (35.6). The ordering of the two routes is therefore a property of the teacher. A fixed 12k endpoint would have detected the elbow body-model benefit and missed the finger's.

Budget. The body model saw $3.3 \times$ (elbow) to $6.7 \times$ (finger) more transitions than T1. Rebuilding the finger model from exactly 30k babble transitions changes its student's 12k endpoint by $+2.75 [-12.77, +18.27]$ mm, and a 30k elbow model changes it by $+1.55 [-0.87, +3.97]$ mrad, neither detected. T2 is not matched to anything: it is the product of 2×10^5 babble transitions and 100k rewarded steps.

B. Withdrawing the teacher: what the cost is made of

At one withdrawal point. Table III decomposes the withdrawal cost at $t_w = 8000$ on both bodies. The immediate dip is the loss term: on the elbow $+40.22 [+19.96, +60.48]$ of a total $+45.17 [+24.58, +65.76]$ mrad, with the teacher component $+4.95 [-1.24, +11.15]$ not detected; on the finger $+46.22 [+30.30, +62.15]$ of $+49.20 [+31.79, +66.60]$ mm, with the teacher component $+2.97 [+0.24, +5.70]$. The `randanchor` condition is worse than `selfanchor` on both bodies, so the anchor's content matters: `selfanchor` is not "any loss term" but one pointing where the student already points, which is what a control for the loss term should be. By 100k the two bodies part. On the elbow the total is $+10.29 [+3.96, +16.62]$ and is almost entirely the teacher component, $+8.16 [+2.17, +14.14]$, the loss-term component $+2.13 [-3.64, +7.90]$ no longer detected: what survives training is the loss of a competent teacher's further information, worth about the same as the teacher's whole advantage over no teacher (constant – none $-8.06 [-15.83, -0.28]$), and the withdrawn learner is not distinguishable from one never guided ($+2.23 [-7.68, +12.14]$). On the finger every component is gone by 100k (total $+0.39 [-3.47, +4.25]$ mm) and the withdrawn learner ends $-59.77 [-73.67, -45.88]$ better than one never guided, having kept nearly all of what eight thousand guided steps gave it; only the random anchor leaves a lasting mark ($+78.73 [+63.05, +94.41]$).

Along the attachment axis. The hypothesis predicts that the cost of losing the teacher grows with how long it was attached. We withdrew at every 1,000 steps from 2,000 to 8,000 on both bodies, running `abrupt` and `selfanchor` at each point to 100k with `constant` and `none` shared (Table IV, Fig. 2). At the dip, the teacher component *falls* with attachment on both bodies ($-6.23 [-7.79, -4.68]$ mrad and $-4.65 [-7.05, -2.25]$ mm per 1,000 attached steps): it tracks how much the teacher is still contributing at the moment of withdrawal, not an accumulated reliance. The loss-term component does not fall; on the finger it grows ($+7.07 [+2.95, +11.19]$) and carries the whole growth of the literature's measure (dip of `abrupt` – none, slope $+9.34 [+2.07, +16.61]$), while on the elbow neither slope is detected. At the shortest attachment, $t_w = 2000$, the immediate drop is mostly the teacher's on both bodies; the deleted term takes over as attachment grows. A study measuring the immediate drop alone would report the hypothesis confirmed on the finger. At 100k the picture is the opposite

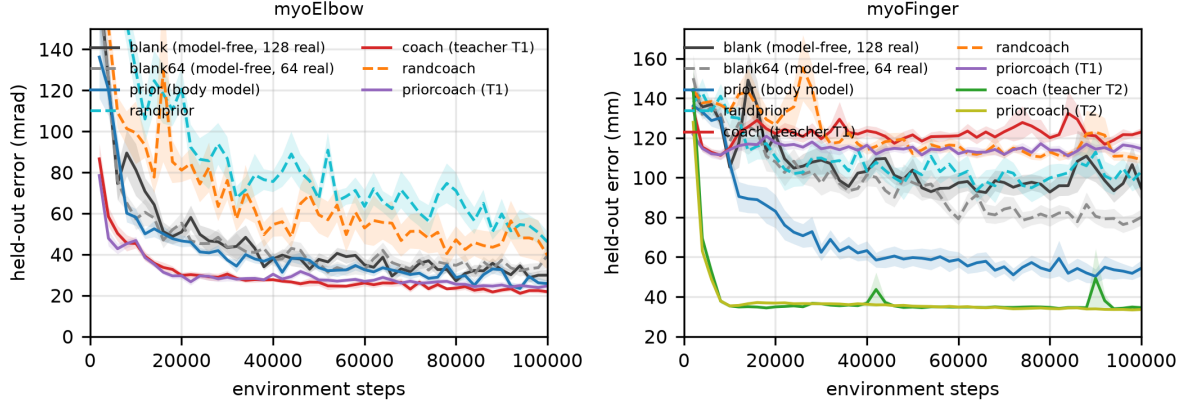


Fig. 1. Held-out error over 100,000 steps on myoElbow (left, mrad; axis clipped at 150) and myoFinger (right, mm): mean across 12 seeds, band ± 1 SE, dashed lines the random controls and blank64. On the finger the teacher conditions are shown with both teachers.

TABLE I

THE CORE CONDITIONS AT 100,000 STEPS, HELD-OUT, $n = 12$. MEANS, THEN PAIRED CONTRASTS WITH 95% INTERVALS; NEGATIVE FAVOURS THE FIRST-NAMED CONDITION; BOLD MARKS AN INTERVAL EXCLUDING ZERO. FINGER TEACHER ROWS ARE GIVEN FOR T1 (NEAR NO-OP) AND T2 (COMPETENT). ‡: EXACT WILCOXON $p = 0.052$.

	myoElbow (mrad)	myoFinger, T1 (mm)	myoFinger, T2 (mm)
blank / blank64	29.87 / 39.55	94.66 / 80.07	
prior / randprior	25.78 / 46.08	54.37 / 103.08	
coach / randcoach	21.81 / 39.50	123.00 / 109.14	34.50 / 109.14
priorcoach	24.63	114.62	33.58
prior-blank	-4.09 [-11.11, +2.94]	-40.29 [-57.95, -22.64]	
prior-blank64	-13.77 [-27.65, +0.11]	-25.70 [-37.48, -13.93]	
blank64-blank	+9.68 [-3.69, +23.06]	-14.59 [-26.00, -3.18]	
prior-randprior	-20.29 [-32.00, -8.59]	-48.71 [-62.32, -35.09]	
randprior-blank	+16.21 [+6.82, +25.59]	+8.41 [-14.17, +30.99]	
coach-blank	-8.06 [-15.83, -0.28]‡	+28.33 [+16.11, +40.56]	-60.17 [-73.51, -46.82]
coach-prior	-3.97 [-8.47, +0.53]	+68.63 [+59.19, +78.07]	-19.87 [-28.20, -11.55]
coach-randcoach	-17.69 [-33.70, -1.68]‡	+13.86 [+7.62, +20.09]	-74.64 [-79.40, -69.89]
priorcoach-coach	+2.81 [-1.02, +6.65]	-8.38 [-15.57, -1.19]	-0.92 [-2.58, +0.74]
coachT2-coachT1		-88.50 [-93.12, -83.88]	

TABLE II

JOINT-COUNT SERIES: FOUR PLANAR MUSCLE ARMS AT 100,000 STEPS, HELD-OUT, IN MM. MEANS, THEN PAIRED CONTRASTS WITH 95% INTERVALS ($df = n - 1$); BOLD MARKS AN INTERVAL EXCLUDING ZERO.

	1 joint ($n=12$)	2 joints ($n=12$)	3 joints ($n=24$)	4 joints ($n=24$)
blank	92.3	34.7	164.4	206.4
blank64	93.4	39.3	152.5	144.0
prior	83.4	26.9	140.7	106.3
randprior	96.7	56.3	292.8	464.2
prior-blank	-8.87 [-14.95, -2.79]	-7.85 [-13.68, -2.01]	-23.76 [-54.70, +7.17]	-100.18 [-156.90, -43.46]
as % of blank	-10%	-23%	-14%	-49%
prior-blank64	-9.93 [-18.36, -1.51]	-12.40 [-18.86, -5.94]	-11.81 [-57.77, +34.15]	-37.76 [-72.54, -2.98]
blank64-blank	+1.07 [-6.13, +8.26]	+4.55 [-1.90, +11.00]	-11.95 [-55.30, +31.40]	-62.42 [-129.04, +4.20]
randprior-blank	+4.34 [-3.60, +12.29]	+21.62 [+6.37, +36.87]	+128.34 [+55.33, +201.35]	+257.72 [+177.95, +337.48]
prior-randprior	-13.21 [-22.83, -3.59]	-29.47 [-41.91, -17.03]	-152.10 [-223.91, -80.29]	-357.90 [-420.40, -295.39]

of the hypothesis. On the elbow abrupt – none is not detected at any t_w and has no trend ($+0.50 [-0.37, +1.37]$); the total is detected at every t_w , $+7.6$ to $+17.0$ mrad, flat; the loss-term component is never detected and the teacher component is detected at four of seven. On the finger the teacher component falls from $+53.28 [+48.14, +58.42]$ mm at $t_w = 2000$ to $+0.18 [-2.32, +2.68]$ at $t_w = 8000$

(slope $-7.94 [-9.23, -6.65]$), the loss-term component is not detected at any t_w , and the withdrawn learner is better than one never guided at six of seven durations, by more the longer the teacher was attached (slope $-8.44 [-9.89, -6.99]$). The one transient in the elbow's favour of the hypothesis is at 50k, where the total's trend is $+1.18 [+0.18, +2.17]$ mrad per 1,000 steps; it is gone by 100k. Two conditions of the

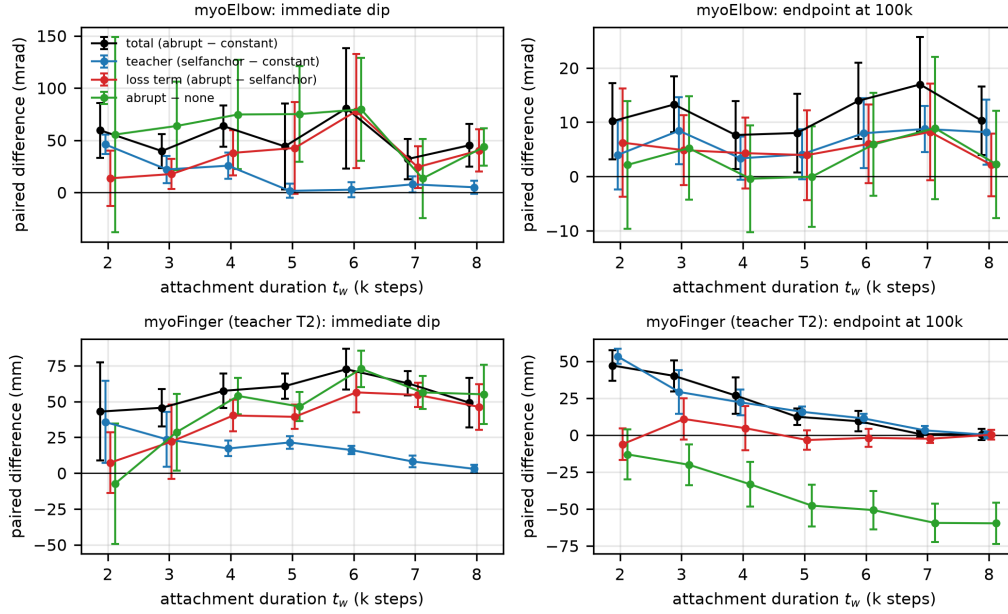


Fig. 2. Table IV drawn: the withdrawal-cost components against attachment duration on myoElbow (top, mrad) and myoFinger with teacher T2 (bottom, mm), at the immediate dip (left) and at 100k (right); paired means with 95% intervals, $n = 12$.

TABLE III

TEACHER WITHDRAWAL DECOMPOSED AT $t_w = 8000$, $n = 12$: DIP, THEN THE SAME CONTRASTS AT 12K AND AT 100K. ROWS 1–3 ARE THE TOTAL, THE TEACHER COMPONENT AND THE LOSS-TERM COMPONENT. * MARKS AN INTERVAL EXCLUDING ZERO. ELBOW IN MRAD (TEACHER T1), FINGER IN MM (TEACHER T2). ‡: EXACT WILCOXON $p = 0.052$.

Contrast	myoElbow			myoFinger, T2		
	dip	12k	100k	dip	12k	100k
abrupt – constant	+45.17*	+16.00*	+10.29*	+49.20*	+27.34*	+0.39
selfanchor – constant	+4.95	+9.42*	+8.16*	+2.97*	+3.61*	+0.18
abrupt – selfanchor	+40.22*	+6.59	+2.13	+46.22*	+23.72*	+0.22
randanchor – selfanchor	+47.44*	+20.66*	+3.26	+70.47*	+85.93*	+78.73*
abrupt – none	+43.64*	–15.60	+2.23	+55.03*	–55.43*	–59.77*
constant – none	–31.61*	–8.06*‡		–82.77*	–60.17*	

sweep are still moving at 100k, the elbow selfanchor at $t_w = 6000$ ($-2.88 [-4.60, -1.17]$) and the finger abrupt at $t_w = 8000$ ($-4.75 [-8.97, -0.52]$), both downward.

C. When guidance replaces the learner’s own signal

The additive coach is the wrong model of physical guidance, which substitutes for the learner’s own error correction rather than supplementing it. The replacing regime is the right one, and it is also what reward-free distillation does in current legged-locomotion pipelines [19], [20]. We ran it on both bodies at $t_w \in \{2000, 4000, 6000, 8000\}$ with constant, abrupt, selfanchor and gateon to 100k (Table V). The bodies differ in what the collapse at withdrawal is and agree that it does not last.

On the elbow, four thousand steps after withdrawal the hypothesis appears confirmed: the cost of losing the teacher is large at every t_w and grows with attachment (selfanchor – constant slope $+5.77 [+1.30, +10.25]$), an order of magnitude above the additive regime. The gateon condition dissolves

it: keeping the real teacher while switching on the student’s own gradient produces the same collapse, seed by seed (selfanchor – gateon within 5.0 mrad at every t_w , no slope). What grows is the cost of first activating the learner’s own policy gradient after a cloning-only phase, the known collapse when a cloned policy is fine-tuned by off-policy reinforcement [30], [31], [32]. On the finger the collapse is the deleted term instead, as in the additive regime: plain SAC after withdrawal is 59 to 93 mm above the never-withdrawn student four thousand steps later, while keeping the teacher costs 3 to 7 mm, as a retained cloning term is reported to do [33], and nothing grows with attachment (abrupt – constant slope $-2.19 [-5.78, +1.40]$).

By 100k the two bodies agree. Plain SAC after withdrawal ends not distinguishably from a learner never guided on the elbow (abrupt – none $+1.0$ to $+3.2$, none detected, slope $+0.22 [-0.65, +1.09]$) and 38 to 59 mm ahead of it on the finger, by more the longer the teacher was attached (slope $-3.20 [-4.91, -1.50]$); it is 4.0 to 6.2 mrad above the never-withdrawn student on the elbow (detected at $t_w \geq 4000$) and above it on the finger only after the earliest withdrawal. Keeping the teacher leaves the learner 2 to 5 mrad or mm ahead of the never-withdrawn student on both bodies. The pure-distillation student is level with model-free learning on the elbow ($-3.02 [-9.59, +3.55]$) and far ahead of it on the finger ($-58.32 [-71.38, -45.26]$). One condition is still moving at 100k, the elbow gateon at $t_w = 6000$ ($-2.75 [-5.16, -0.34]$), downward. After a cloning-only phase the student’s snapshot approximates the teacher, so selfanchor cannot price the teacher’s information in this regime; on the finger a snapshot taken at $t_w = 2000$, before the student had cloned its teacher, is the one lasting cost ($+20.72 [+14.50, +26.94]$ against constant).

TABLE IV

TEACHER-WITHDRAWAL COMPONENTS AGAINST ATTACHMENT DURATION t_w : THE IMMEDIATE DIP, AND THE ENDPOINT AT 100K. CELLS ARE PAIRED MEANS, $n = 12$, * AN INTERVAL EXCLUDING ZERO; SLOPE: PER-SEED OLS OVER t_w , PAIRED- t INTERVAL, PER 1,000 ATTACHED STEPS. THE DIP OF ABRUPT — NONE IS THE ABRUPT LEARNER'S DIP MINUS THE NEVER-GUIDED LEARNER'S CHANGE OVER THE SAME 1,000 STEPS.

		2k	3k	4k	5k	6k	7k	8k	slope per 1k
<i>myoElbow (mrad)</i>									
dip	teacher	+46.0*	+22.0*	+25.6*	+1.5	+2.7	+7.8	+5.0	-6.23 [-7.79, -4.68]*
	loss term	+13.5	+17.6*	+37.9*	+42.6	+77.9*	+24.3*	+40.2*	+4.76 [-0.13, +9.65]
	total	+59.5*	+39.7*	+63.5*	+44.1*	+80.6*	+32.1*	+45.2*	-1.47 [-5.79, +2.84]
	abrupt — none	+55.4	+63.7*	+74.7*	+75.0*	+79.6*	+13.4	+43.6*	-4.68 [-14.80, +5.44]
100k	teacher	+4.0	+8.4*	+3.3	+4.1	+8.0*	+8.8*	+8.2*	+0.64 [-0.68, +1.96]
	loss term	+6.2	+4.8	+4.3	+3.9	+6.0	+8.2	+2.1	-0.14 [-1.69, +1.41]
	total	+10.2*	+13.3*	+7.6*	+8.0*	+14.0*	+17.0*	+10.3*	+0.50 [-0.37, +1.37]
	abrupt — none	+2.1	+5.2	-0.4	-0.0	+5.9	+8.9	+2.2	+0.50 [-0.37, +1.37]
<i>myoFinger (mm), teacher T2</i>									
dip	teacher	+35.7*	+23.6*	+17.2*	+21.4*	+16.1*	+8.1*	+3.0*	-4.65 [-7.05, -2.25]*
	loss term	+7.3	+22.0	+40.2*	+39.3*	+56.4*	+54.6*	+46.2*	+7.07 [+2.95, +11.19]*
	total	+43.0*	+45.6*	+57.5*	+60.7*	+72.5*	+62.7*	+49.2*	+2.42 [-2.41, +7.26]
	abrupt — none	-7.3	+28.6*	+53.9*	+46.5*	+72.8*	+56.4*	+55.0*	+9.34 [+2.07, +16.61]*
100k	teacher	+53.3*	+29.2*	+22.2*	+15.7*	+11.3*	+3.2*	+0.2	-7.94 [-9.23, -6.65]*
	loss term	-6.2	+10.9	+4.6	-3.4	-1.9	-2.5	+0.2	-0.50 [-1.61, +0.61]
	total	+47.1*	+40.1*	+26.8*	+12.4*	+9.4*	+0.7	+0.4	-8.44 [-9.89, -6.99]*
	abrupt — none	-13.1	-20.1*	-33.4*	-47.8*	-50.8*	-59.5*	-59.8*	-8.44 [-9.89, -6.99]*

TABLE V

REPLACING REGIME: ERROR 4,000 STEPS AFTER WITHDRAWAL (POST-4K) AND AT 100K, AT EACH t_w , WITH PER-SEED SLOPES PER 1,000 ATTACHED STEPS; PAIRED MEANS, $n = 12$, * AN INTERVAL EXCLUDING ZERO.

	post-4k					endpoint 100k				
	2k	4k	6k	8k	slope	2k	4k	6k	8k	slope
<i>myoElbow (mrad), teacher T1</i>										
selfanchor — constant	+28.1*	+64.8*	+88.1*	+58.8*	+5.77 [+1.30, +10.25]*	+3.4	-1.1	-4.3*	-3.9*	-1.26 [-1.97, -0.54]*
gateon — constant	+25.0*	+69.8*	+84.7*	+59.4*	+5.90 [+0.26, +11.54]*	-3.9*	-2.7*	-4.3*	-4.8*	-0.22 [-0.73, +0.29]
selfanchor — gateon	+3.1	-5.0	+3.4	-0.6	-0.13 [-3.44, +3.19]	+7.3*	+1.6	+0.0	+0.9	-1.04 [-1.88, -0.19]*
abrupt — selfanchor	+51.9*	+136.9*	+96.8	+94.1*	+4.32 [-4.60, +13.24]	+0.6	+7.3*	+10.2*	+9.5*	+1.48 [+0.26, +2.69]*
abrupt — constant	+80.0*	+201.7*	+184.9*	+152.9*	+10.10 [+0.52, +19.68]*	+4.0	+6.2*	+5.9*	+5.6*	+0.22 [-0.65, +1.09]
abrupt — none	+34.6*	+140.3*	+131.3*	+110.1*	+10.87 [-0.11, +21.85]	+1.0	+3.2	+2.9	+2.5	+0.22 [-0.65, +1.09]
<i>myoFinger (mm), teacher T2</i>										
selfanchor — constant	+46.8*	+5.7*	+5.5*	+8.3*	-5.79 [-7.00, -4.58]*	+20.7*	-1.1	-3.4*	-4.2*	-3.85 [-4.92, -2.78]*
gateon — constant	+6.7*	+3.0*	+7.3	+6.7*	+0.21 [-0.92, +1.35]	-2.0*	-1.8*	-2.0*	-2.2*	-0.03 [-0.26, +0.19]
selfanchor — gateon	+40.1*	+2.7*	-1.7	+1.6*	-6.00 [-7.18, -4.83]*	+22.8*	+0.6	-1.4*	-2.0*	-3.82 [-4.85, -2.78]*
abrupt — selfanchor	+46.4*	+53.6*	+62.4*	+67.5*	+3.60 [-0.09, +7.30]	-0.1	+3.4	+4.8*	+3.8*	+0.65 [-0.95, +2.24]
abrupt — constant	+93.2*	+59.4*	+67.9*	+75.8*	-2.19 [-5.78, +1.40]	+20.7*	+2.3	+1.5	-0.4	-3.20 [-4.91, -1.50]*
abrupt — none	-9.1	-35.3*	-3.7	-8.4	+1.67 [-4.81, +8.15]	-37.7*	-56.0*	-56.9*	-58.7*	-3.20 [-4.91, -1.50]*

TABLE VI

WITHDRAWAL OF THE BODY MODEL AT 8K, THE SAME RUNS SCORED AT 12K AND AT 100K, $n = 12$. PAIRED MEANS; POSITIVE MEANS WITHDRAWAL MADE THE CONDITION WORSE; * MARKS AN INTERVAL EXCLUDING ZERO; †: SINGLE-FACTOR. THE LAST ROWS ARE THE LITERATURE'S MEASURE, THE WITHDRAWN LEARNER AGAINST ONE THAT NEVER HAD A MODEL. MEANS AT 12K / 100K, ELBOW: KEEP 50.02 / 25.78, SOFT 52.11 / 33.18, PURGE 57.19 / 31.12, MATCHED 72.13 / 30.84; FINGER: KEEP 90.50 / 54.37, SOFT 88.11 / 58.50, PURGE 104.69 / 74.52, MATCHED 100.48 / 69.68.

	myoElbow (mrad)		myoFinger (mm)	
	12k	100k	12k	100k
purge-keep	+7.17	+5.34	+14.19	+20.15*
soft-keep†	+2.09	+7.40*	-2.39	+4.13
matched-keep†	+22.11*	+5.05	+9.97	+15.31*
soft-matched†	-20.02*	+2.35	-12.36	-11.18
purge-matched	-14.94	+0.28	+4.22	+4.84
soft-blank	-	+3.32	-	-36.17*
purge-blank	-	+1.25	-	-20.15*
matched-blank	-	+0.97	-	-24.99*

D. Withdrawing the body model

Table VI withdraws the body model at 8k. Measured as the standard protocol measures it, the withdrawal has no detectable cost at 12k: purge — keep is +7.17 [-10.95, +25.29]. With distinct real transitions per update held at 64 the same withdrawal is significantly costly, matched — keep +22.11 [+4.94, +39.28], because the strict withdrawal doubles the agent's real-data gradient signal at the moment it loses the model and that gain offsets part of the loss; the direct contrast purge — matched is -14.94 [-44.62, +14.73], not detected, so we put no number on the offset. With real-data content equal, keeping the stale imagined transitions is worth -20.02 [-37.34, -2.70] (soft — matched): by 8000 steps what the frozen model is worth on this body is carried by the imagined transitions already in the buffer. By 100k the single-factor cost has shrunk to +5.05 [-0.98, +11.08] and is not detected; the one contrast detected runs the other way from 12k, the stale buffer now the worst of the four (+7.40 [+2.72, +12.08]); and each withdrawn condition ends where a learner that never had a model does (soft, purge, matched against blank: +3.32 [-3.73, +10.36],

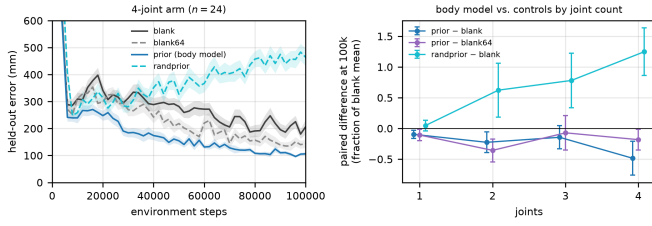


Fig. 3. Joint-count series. Left: the four-joint arm’s conditions over training (mean, ± 1 SE, $n = 24$). Right: paired contrasts at 100k as a fraction of each arm’s mean model-free error, with 95% intervals.

+1.25 [−7.27, +9.78], +0.97 [−6.59, +8.53]). The finger runs the other way in time. Neither protocol detects a cost at 12k (purge – keep +14.19 [−0.95, +29.33] mm; matched – keep +9.97 [−1.50, +21.45]), and both do at 100k (+20.15 [+11.64, +28.66] and +15.31 [+5.70, +24.91]): on the body where the model keeps paying, a learner that loses it at 8k forgoes the imagination it would have had for the next 92,000 steps. It is still far better off than a learner that never had a model: every withdrawn condition ends below *blank*, the matched one by −24.99 [−44.86, −5.11], keeping 62% of the advantage of never losing the model (Table I). The finger runs reproduce the 12k runs seed for seed at 2k–12k.

V. DISCUSSION

What dependence looks like when it can be taken apart.

Informationless stand-ins for guidance are rare in human studies, so dependence is measured as the withdrawn learner against one never guided, and growth with attachment is read as accumulated reliance. The finger reproduces that growth at the immediate drop, and the self-anchor attributes it: the teacher component falls with attachment on both bodies, because it tracks what the teacher was still contributing, and the deleted term’s shock is what grows. Whether an optimisation event that grows with attachment should be called dependence is a matter of definition; what remains at convergence is not. There the withdrawn learner is never detectably worse than one never guided, and on the finger it is better by more the longer it was guided, the reverse of the hypothesis. On the elbow what persists is 3 to 9 mrad of a competent teacher’s further information, of the size of the teacher’s whole advantage and flat in attachment. The body model behaves alike: its withdrawal leaves the finger learner 15 mm short of one that kept it and 25 mm ahead of one that never had it. In the replacing regime on the elbow the cost that grows with attachment belongs to an actor that has never been moved by its critic, the hypothesis’ mechanism in a form the additive regime cannot show; on the finger the collapse is the deleted term and does not grow; on both it is gone by convergence. Our evaluation is always greedy and without the teacher, so every checkpoint is a no-guidance transfer test, and our dip follows continued practice, the analogue of a delayed retention test with practice in between rather than the no-practice test the guidance literature favours [10]. Human studies do find withdrawal costs at retention after fixed practice [6], [71], and a field study with our yaststick found students whose tutor

was withdrawn below students who never had one [72]. Our claim is about the state after convergence, where a weaned learner can end above a from-scratch one [22] and a regulariser removed after the early phase costs little [73], though early experience can also leave lasting marks on a deep RL learner [74]. Here it did not.

What the priors are worth. A frozen babble model is worth something on four of six bodies and harmful on none, and what it is worth is its content: a random model in the same machinery harms by more the more joints there are, a series the model’s own benefit does not follow, so a harder body makes a wrong model cost more rather than a right one worth more. A teacher is worth its competence: with a near-no-op and a competent teacher the same body, seeds and loss give the worst finger condition and one level with the best, so the two routes cannot be ordered on one body with one teacher.

Measurement. A withdrawal that changes two things at once cannot be read as the cost of the thing withdrawn: the body model needed a matched-replay condition, the teacher a self-anchor, and a cloning-only learner a *gateon* condition, since a distillation learner with no reward term [19], [20] freezes when its coefficient is zeroed. Fading schedules deserve the same decomposition, since a taper may reduce the shock rather than the dependence. And an endpoint fixed in advance sat inside a transient on both MyoSuite bodies; the remedy is a per-condition plateau check, which here leaves five of 97 conditions moving.

VI. LIMITATIONS

The study is single-task reach in simulation, on deterministic, fully observed bodies, which favour a frozen forward model; the one-joint bodies’ held-out targets interpolate between the training targets. The finger’s first teacher had drifted to within 3.2 mm of a zero-activation policy, so its results concern distilling toward a near-no-op target. The self-anchor carries what the student had absorbed by withdrawal, so the fall of the teacher component with attachment is partly by construction; the convergence claim rests on abrupt – none, which needs no anchor. The body-model condition is handed the analytic reward at imagined states. Withdrawal is a single step, attachment was varied over seven points, and contrasts were not pre-registered. No physical robot is involved.

VII. CONCLUSION

Two routes to a motor prior were compared under one learner on six muscle-driven bodies, to a checked plateau, with adversarial controls and single-factor withdrawals. What a teacher still contributes at withdrawal falls with attachment; what grows, where anything does, is the shock of deleting its loss term. At convergence a learner that lost a teacher or a body model is never behind one that never had it; what can remain is a shortfall against keeping the prior. What the priors were worth is bound to their content: a babble model helps on four of six bodies, a random one harms by more the more joints there are, and a teacher is worth its competence.

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